

FRESHMAN SEMINAR · RECITATION HOMEWORK

Homework 2: Big-Idea Hunting

Name: _____

Date: _____

Instructions: About 20 minutes. Answer in your own words — if your answer means the same thing as the key, it counts. Part C is the main event: you get to be the professor and write a quiz question. Total: **21 points**.

Part A — Boil it down

From NICU Tutor: Drinking milk seems like the easiest thing a baby does. It's actually a stunt. To feed safely, a baby must suck, then swallow, then breathe — in perfect rhythm, over and over, without ever letting milk slip down the wrong pipe into the lungs. That rhythm is a brain skill, and it isn't wired up until about 32 to 34 weeks of development. A baby born at 26 weeks simply doesn't have the wiring yet. Feed that baby from a bottle and milk would end up in its lungs. So the NICU delivers milk another way: a soft tube from the nose to the stomach, with a pump pushing the milk in slowly.

1. Write the **ONE** biggest idea of that passage in your own words — one line. Then, underneath, write one detail you chose to leave out. (3 pts)

From this morning's Chemistry: Look back at the ruler. Blood sits at 7.35 to 7.45. That window is one-tenth of one step wide. Your stomach is happy at 2; your skin is fine at 5; but blood that drifts to 7.2 makes a NICU team move fast, and blood at 6.9 or 7.8 is usually fatal. Why so picky? Because of proteins. A protein works because of its exact folded shape, and loose hydrogens grab onto proteins and warp them. A little too much acid and the enzymes slow down, the heart muscle squeezes more weakly, and hemoglobin drops its oxygen too early.

2. Same again: the **ONE** big idea, one line, your words. Then one detail you left out. (3 pts)

Part B — Idea or detail?

3. Six sentences from this week's Monday classes. Mark each one I (big idea) or D (detail), and say which test you used (so what? / forget it / repeat). (6 pts)

Sentence	I or D?	Which test?
Sørensen invented pH in 1909 at a brewery.		
Each pH step is ten times more acidic.		
The heart is two pumps: right side to the lungs, left side to the body.		
A newborn's heart is about the size of a walnut.		
Testing yourself works better than rereading.		
Leonardo wrote in mirror writing.		

Part C — You're the professor

4. Write ONE multiple-choice quiz question about something from today's Chemistry or Biology. Include the question, the right answer, and TWO wrong answers that sound reasonable (no jokes, and about the same length as the right one). Remember the rule: someone should have to actually know the material. (4 pts)

Part D — Flashbacks

5. From NICU Tutor: what does the isolette's skin probe do? (1 pt)

6. From last week's Seminar: what are office hours? (1 pt)

Part E — Think like a student

7. A friend says: “I reread the chapter three times and still failed the quiz.” Using today's class, tell them what to do instead – and why it works better. (3 pts)

Score: _____ / 21 points

Professor's comment:
